



RAMCO INSTITUTE OF TECHNOLOGY

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NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year: 2025-2026(Odd Semester)

INNOVATIVE TEACHING PRACTICES

Degree, Semester & Branch: V Semester B.E. CSE

Course Code & Title: CCS375 Web Technologies

Name of the Faculty member: Dr. S. Poornam

Date & Time: 23.10.2025 & 04:30 P.M

Topic: Conditional Directives in Angular JS

Course Outcome: CO5

Type of Activity: Activity-Based Learning

Justification

This innovative classroom activity introduces students to the conditional directives in AngularJS such as ng-if, ng-show, and ng-hide through a real-time coding simulation. By enabling students to alter UI visibility dynamically based on logical conditions, the activity bridges theoretical Angular directive usage with practical application. The approach promotes active experimentation, real-time debugging, and collaborative learning, helping students internalize the difference between structural and attribute-based conditional rendering.

Time Allotted for the Activity: 20 Minutes

Details of the Implementation

- ❖ Students were divided into **five small groups**, each provided with a template AngularJS code snippet containing UI elements.
- ❖ Each group's task was to implement conditional display mechanisms using AngularJS directives:
 - **ng-if** – to render elements conditionally
 - **ng-show** – to toggle visibility (true/false)
 - **ng-hide** – to hide elements based on Boolean conditions

Sample requirement provided to groups: Display the text “Eligible for Drive” only if the student mark is greater than or equal to 75; otherwise show “Not Eligible”.

- ❖ Students tested with various input values and observed directive behavior in real-time.
- ❖ Each group demonstrated how their code output changed dynamically.
- ❖ Instructor clarified the difference between:
 - ❖ Component **creation** vs **visibility toggling**
 - ❖ Effect on **DOM manipulation & rendering cost**

Learning Outcome:

- ❖ Students gained the ability to apply conditional directives effectively in AngularJS applications.
- ❖ Understood differences between structural (ng-if) and behavioral (ng-show / ng-hide) directives.
- ❖ Enhanced debugging skills by testing UI changes interactively.
- ❖ Improved collaborative coding and problem-solving skills.

CO – PO / PSO Mapping

CO	PO1	PO2	PO3	PO4	PSO1
CO1	2	2	3	3	2

(1 – Low, 2 – Moderate, 3 – High)

Innovative practice	PO1	PO2	PO3	PO4	PSO1
	2	2	3	3	2
Justification for correlation	Students applied logical reasoning in implementing conditional checks.	Analytical thinking developed through evaluating directive outcomes.	Practiced design and implementation of dynamic UI behavior in AngularJS.	Understood the influence of UI state control on user interaction and application performance.	Gained practical exposure to frontend behavior in real-world web applications.

PO/PSO mapped: PO1, PO2, PO3, PO4, PSO1

Images / Screenshot of the practice:



Reflective Critique:

- ❖ Students expressed that hands-on experimentation helped them clearly differentiate among conditional directives.
- ❖ Real-time testing improved understanding of dynamic UI manipulation.
- ❖ Team collaboration enabled faster debugging and idea sharing.
- ❖ Visualization of output made conceptual learning more memorable.

Benefits of the Practice:

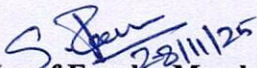
- ❖ Improved problem-solving and practical application of theory.
- ❖ Enhanced retention through visually observable output variations.
- ❖ Increased student engagement and active participation.
- ❖ Strengthened coding confidence for real client-driven UI interactions.

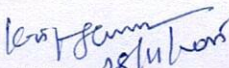
Challenges Faced in Implementation:

- ❖ Initial difficulty in understanding DOM manipulation differences across directives.
- ❖ Some teams required support with syntax errors in AngularJS.
- ❖ Time constraints limited advanced exploration like nested conditions.

References

- <https://www.riipen.com/blog/innovative-teaching-methods>
- https://bssspublications.com/PublishedPaper/Publish_245.pdf
- <https://piogroup.net/blog/20-innovative-teaching-methods-with-examples-how-to-implement-in-education-process>
- <https://peer.asee.org/innovative-activities-to-teach-computer-science-concepts-inside-the-classroom-and-at-outreach-events.pdf>


Signature of Faculty Member


HOD